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10.1148/rycan.240161

Appendix S1

Implementation of Baseline Methods

Random forest (RF) models were implemented in accordance to commonplace radiomic analysis pipelines consisting of feature extraction, feature selection, model training, and model evaluation. The radiomic features used as input to models were the min-max normalized radiomic features from the gross target volume (GTV) region on patient CT images. As described in the Materials and Methods section, radiomic features were first selected using the minimum redundancy maximum relevance (MRMR) algorithm. Specifically, 20 features were initially selected. Then, every combination of one to 20 features from this subset were used to train RF models using the training dataset. Each combination was also trained using either undersampling, oversampling, or a combination of the two to combat class imbalance. Each of these models was evaluated on the held-out validation set, and the top performing model was selected based upon area under the receiver operating characteristic curve (AUC) for local-regional (LR) or distant metastasis (DM), thus determining the number of features and samplings strategy used to generate predictions on the test set. The clinical baseline model was implemented in the same fashion, replacing radiomic features with one-hot encoded clinical features.

The Diamant et al (14) baseline convolutional neural network (CNN) model was implemented in accordance with the details provided in Diamant et al (14). The authors did provide code for their model but did not provide code to process native CT volumes into the single section images described in their paper. As a result, we attempted to replicate the process as faithfully to the original work as possible. However, as can be observed in the Results section, performance using this CNN approach did not achieve satisfactory results. It is not clear what may be the cause of these findings, though we do note that the data domain of our work is significantly different in scope (n = 3434 vs n = 300) to that of the original work. Further, our use of independent validation and testing datasets might account for the lower performance by avoiding overfitting that might occur in the original study's framework, in which only training and validation datasets were used. These findings mirror those by Mateus et al (15) who also reported an inability to replicate the findings of Diamant et al (14).

The Mateus et al (15) baseline CNN was implemented using the provided code. Slight alterations were made to accommodate the larger scope of this dataset including adjustments to enable the inclusion of lower resolution CT scans present for some patients.

For both CNN approaches, a hyperparameter search was performed to enable a fair comparison to RadGraph. Hyperparameter settings were determined on the validation set for both CNN and RadGraph models.

Clinical categorical variables were one-hot encoded prior to input into each model, and quantitative categorical variables were normalized so that their values remained between 0 and 1.

It should be noted that sensitivities and specificities presented in Tables S4 and S5 were calculated with a typical threshold value of 0.5. As a result, we observe sensitivities and specificities of 1.00 and 0.00 in certain cases. We present AUC in the main paper as a more

complete metric of model performance than sensitivity or specificity. While sensitivity and specificity are metrics that can be tuned for a given model to achieve a desired balance specific to a given classification task, AUC is a constant metric of model performance which is invariant to different decision thresholds.

Table S1

CT Imaging Parameters

Dataset	Peak Voltage (kVp) Median (Range)	Exposure Time (mAs) Median (Range)	Spacing (mm) Median (Range)	Section Thickness (mm) Median (Range)
RADCURE (*n* = 2487)	120 (120–120)	300 (21–13614)	0.976 (0.672–1.195)	2.00 (2.00–2.5)
HNPET (*n* = 288)	120 (120–140)	210 (5–13142)	1.020 (0.605–1.953)	3.00 (1.50–3.75)
MDACC (*n* = 535)	120 (100–140)	20 (0–21103)	0.488 (0.340–0.977)	1.25 (1.00–5.00)
HN1 (*n* = 124)	Not Provided	Not Provided	0.977 (0.977–1.088)	3.00 (1.50–9.00)

Table S2

Network Architecture and Hyperparameters

Hyperparameters	Values Tested	LR Model Value	DM Model Value	Description
Number of GAT Layers	1, 2, 3, 5	5	3	A GAT layer consists of a single pass of multiheaded graph attention between neighboring nodes as described in Veličković et al (16).
Hidden Feature Dimension	4, 8, 16	16	8	The dimensions of intermediate node features after each GAT layer.
Number of Attention Heads	1, 2, 4, 8	4	4	The number of parallel attention values calculated during each GAT layer.
Dropout Probability	(0, 0.8)	0.67	0.89	Dropout is commonly applied to regularize deep learning networks. For GATs, dropout masks attention weights in GAT layers.
Learning Rate	(5e-5, 5e-3)	4e-4	3e-4	Adjusts the speed at which model weights are updated during training.
Optimizer	Adam, SGD	Adam	Adam	Adjusts model parameters based upon the loss function and learning rate.
Weight Decay	(1.0e-6, 1.0e-3)	7e-4	8e-4	A regularization technique that ensures model weights do not grow too large during extended training.
Neighbors	5, 20, 50, 100	20	20	The number of nodes retained in radiomic graph creation among the ~ 100 supervoxels identified per patient.
Sampling Strategy	Under, Over, Intermediate	Intermediate	Over	Approaches that either leverage multiple copies of the minority class, only a subset of the majority class, or some combination of the two to combat class imbalance when training models.
Learning Rate Warmup Steps	5, 10	10	5	It has been found that slowly increasing the learning rate during training rather than commencing with the full desired learning rate can improve training stability.
Batch Size	64, 128, 256	64	64	The number of patient graphs trained in each batch.

Table S3

Radiomic Features Used for LR and DM Prediction

Local-regional Recurrence	Distant Metastasis
1. GLSZM Gray Level Non-Uniformity	1. NGTDM Coarseness
2. GLCM Inverse Difference	2. First Order Mean Intensity
3. GLCM Inverse Difference Normalized	3. GLDM Large Dependence Low Gray Level Emphasis
4. GLSZM Zone Entropy	4. GLCM Cluster Prominence
	5. GLSZM Low Gray Level Zone Emphasis
	6. GLSZM Small Area Emphasis

Note.—GLSZM = gray level size zone matrix, GLCM = gray level co-occurrence matrix, NGTDM = neighboring gray tone difference matrix, GLDM = gray level dependence matrix.

Table S4

List of All Radiomic Features Studied

First Order Features	Gray Level Run Length Matrix Features
• Entropy	• Emphasis
• Entropy	• Non-Uniformity
• Intensity Statistics	• Run Percentage
Shape Features	• Variance
• Volume	• Entropy
• Surface Area	Neighboring Gray Tone Difference Matrix Features
• Sphericity	• Coarseness
• Compactness	• Contrast
• Diameter	• Busyness
• Elongation	• Complexity
• Axis Lengths	• Strength
Gray Level Co-occurrence Matrix Features	Gray Level Dependence Matrix Features
• Autocorrelation	• Emphasis
• Joint Average	• Non-Uniformity
• Prominence	• Variance
• Shade	• Entropy
• Tendency	Gray Level Size Zone Matrix Features
• Contrast	• Emphasis
• Correlation	• Non-Uniformity
• Difference Average	• Zone Percentage
• Difference Entropy	• Variance
• Difference Variance	• Entropy
• Joint Energy	
• Joint Entropy	
• Informational Measure of Correlation	
• Inverse Difference	
• Maximum Probability	
• Sum Average	
• Sum Entropy	

Table S5

Full Metrics for LR Prediction

	RADCURE			HNPET			MDACC			HN1		
	AUC	Sens	Spec	AUC	Sens	Spec	AUC	Sens	Spec	AUC	Sens	Spec
Clinical Baseline	0.73 ± 0.04	54% (8%)	78% (2%)	0.68 ± 0.05	93% (5%)	37% (3%)	0.57 ± 0.05	56% (8%)	49% (3%)	0.66 ± 0.05	63% (9%)	62% (5%)
Traditional Radiomics (imaging only)	0.59 ± 0.04	63% (7%)	53% (2%)	0.54 ± 0.06	43% (10%)	63% (3%)	0.63 ± 0.04	51% (7%)	67% (3%)	0.58 ± 0.06	37% (9%)	65% (5%)
Diamant et al (14) CNN (imaging only)	0.44 ± 0.04	0% (0%)	100% (0%)	0.47 ± 0.07	0% (0%)	100% (0%)	0.47 ± 0.05	2% (2%)	99% (1%)	0.47 ± 0.07	0% (0%)	100% (0%)
Mateus et al (15) CNN (imaging only)	0.64 ± 0.04	0% (0%)	100% (0%)	0.59 ± 0.06	0% (0%)	100% (0%)	0.64 ± 0.04	0% (0%)	100% (0%)	0.64 ± 0.06	0% (0%)	100% (0%)
RadGraph (imaging only)	0.65 ± 0.04	71% (6%)	49% (3%)	0.57 ± 0.06	47% (11%)	59% (4%)	0.57 ± 0.05	75% (6%)	26% (2%)	0.66 ± 0.07	47% (6%)	75% (5%)
Traditional Radiomics (imaging and clinical)	0.65 ± 0.04	54% (6%)	62% (2%)	0.65 ± 0.06	37% (10%)	67% (3%)	0.52 ± 0.04	46% (7%)	58% (2%)	0.64 ± 0.06	60% (10%)	60% (6%)
Diamant et al (14) CNN (imaging and clinical)	0.44 ± 0.04	0% (0%)	100% (0%)	0.50 ± 0.06	0% (0%)	100% (0%)	0.46 ± 0.05	0% (0%)	100% (0%)	0.45 ± 0.07	0% (0%)	100% (0%)
Mateus et al (15) CNN (imaging and clinical)	0.81 ± 0.03	0% (0%)	100% (0%)	0.75 ± 0.04	0% (0%)	100% (0%)	0.61 ± 0.05	0% (0%)	100% (0%)	0.80 ± 0.04	0% (0%)	100% (0%)
RadGraph (imaging and clinical)	0.83 ± 0.03	67% (7%)	81% (2%)	0.76 ± 0.04	77% (10%)	66% (3%)	0.60 ± 0.05	41% (6%)	78% (2%)	0.81 ± 0.05	53% (10%)	85% (4%)

Note.—Area under the receiver operating characteristic curve (AUC) values reported as means ± SDs. SD values were calculated via 100-fold boot-strapping of each dataset studied.

Table S6

Full Metrics for DM Prediction

	RADCURE			HNPET			MDACC			HN1		
	AUC	Sens	Spec	AUC	Sens	Spec	AUC	Sens	Spec	AUC	Sens	Spec
Clinical Baseline	0.71 ± 0.04	79% (6%)	57% (3%)	0.59 ± 0.06	96% (4%)	12% (2%)	0.50 ± 0.07	96% (4%)	6% (1%)	0.83 ± 0.07	86% (14%)	63% (6%)
Traditional Radiomics (imaging only)	0.62 ± 0.04	4% (2%)	97% (1%)	0.60 ± 0.07	11% (7%)	99% (1%)	0.60 ± 0.06	4% (5%)	97% (1%)	0.72 ± 0.11	0% (0%)	99% (1%)
Diamant et al (14) CNN (imaging only)	0.39 ± 0.04	0% (0%)	100% (0%)	0.45 ± 0.06	0% (0%)	100% (0%)	0.43 ± 0.07	0% (0%)	100% (0%)	0.25 ± 0.11	0% (0%)	100% (0%)
Mateus et al (15) CNN (imaging only)	0.72 ± 0.04	0% (0%)	100% (0%)	0.57 ± 0.08	0% (0%)	100% (0%)	0.65 ± 0.05	0% (0%)	100% (0%)	0.56 ± 0.09	0% (0%)	100% (0%)
RadGraph (imaging only)	0.73 ± 0.04	79% (6%)	50% (3%)	0.71 ± 0.05	70% (9%)	50% (3%)	0.62 ± 0.07	89% (8%)	18% (2%)	0.90 ± 0.06	100% (10%)	60% (6%)
Traditional Radiomics (imaging and clinical)	0.70 ± 0.03	71% (6%)	63% (3%)	0.61 ± 0.04	100% (0%)	0% (0%)	0.55 ± 0.03	100% (0%)	0% (0%)	0.57 ± 0.12	71% (10%)	51% (5%)
Diamant et al (14) CNN (imaging and clinical)	0.38 ± 0.04	0% (0%)	100% (0%)	0.42 ± 0.00	0% (0%)	100% (0%)	0.43 ± 0.06	0% (0%)	100% (0%)	0.24 ± 0.09	0% (0%)	100% (0%)
Mateus et al (15) CNN (imaging and clinical)	0.79 ± 0.03	0% (0%)	100% (0%)	0.77 ± 0.05	0% (0%)	100% (0%)	0.69 ± 0.05	0% (0%)	100% (0%)	0.87 ± 0.05	0% (0%)	100% (0%)
RadGraph (imaging and clinical)	0.79 ± 0.03	77% (5%)	69% (2%)	0.79 ± 0.05	100% (10%)	0% (3%)	0.64 ± 0.06	92% (6%)	18% (2%)	0.86 ± 0.05	57% (5%)	83% (2%)

Note.—Area under the receiver operating characteristic curve (AUC) values reported as means ± SDs. SD values were calculated via 100-fold boot-strapping of each dataset studied.